

INTERNSHIP REPORT PRESENTATION

# Google AI-ML

## Virtual Internship

AICTE – EduSkills Virtual Internship Program | Supported by Google for Developers

**Param Agarwal**

Roll No. 2410030879 | B.Tech Computer Science and Engineering

June – August 2026



**Param  
Agarwal**

B.Tech – Computer Science & Engineering

## ABOUT THE STUDENT

# Candidate Profile



**Roll Number**

2410030879



**Institute**

IILM University, Greater Noida, U.P.



**Programme**

B.Tech CSE, Batch 2024–2028



**Internship Domain**

Google AI-ML



**Duration**

8 Weeks | June – August 2026

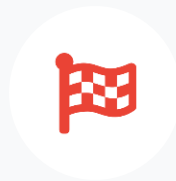
# EduSkills, AICTE & Google for Developers

EduSkills Academy is an organization certified under ISO 9001:2015 and ISO/IEC 20000-1:2018, with a primary focus on skill development in support of the vision of “Nation Building Through Skills.” The organization works in association with the All India Council for Technical Education (AICTE) to facilitate the AICTE–EduSkills Virtual Internship Program through the AICTE National Internship Portal. As part of the Google AI-ML track, EduSkills works in collaboration with Google for Developers to provide a structured, industry-oriented curriculum based on learning pathways, assessments, and skill badges offered through the Google Developer Program.



## ISO Certified

ISO 9001:2015 & ISO/IEC 20000-1:2018 certified organization.



## AICTE Patronage

Run under the AICTE National Internship Portal.



## Google for Developers

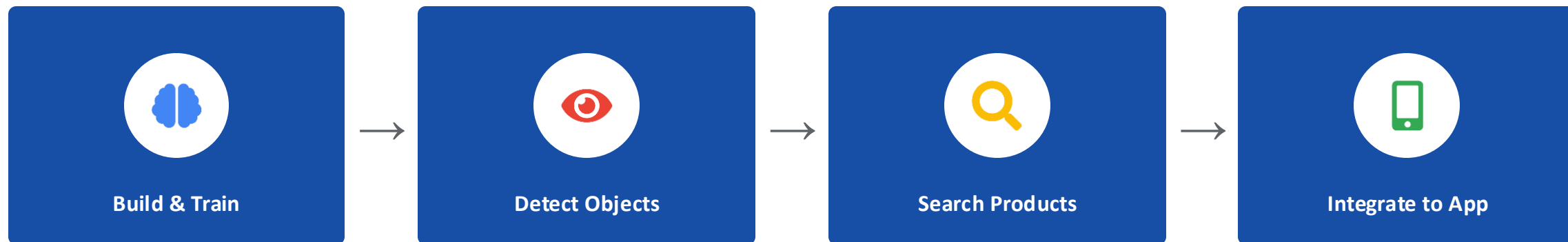
Industry curriculum delivered via the Google Developer Program.

## PROBLEM STATEMENT

# Bridging Theory and Applied Computer Vision

While machine learning and neural network concepts are commonly covered as part of engineering curricula, students often have limited opportunities to gain practical experience in developing and implementing computer vision solutions using industry-standard frameworks such as TensorFlow. Topics including Convolutional Neural Networks (CNNs), on-device object detection, and image classification are frequently addressed primarily from a theoretical perspective. As a result, students may have insufficient practical exposure to the process of converting trained models into functional systems and integrating them into real-world applications.

**This internship addressed that gap — practically learning the full TensorFlow-based workflow:**



# What the Internship Set Out to Achieve



## Build CNNs with TensorFlow

Understand ML from first principles and build convolutional neural networks for image recognition.



## Object Detection

Integrate pretrained detectors and train custom models with TensorFlow Lite & Model Maker.



## Product Image Search

Build a visual, on-device product search feature across static images and live camera feeds.



## Backend Integration

Build backend integration for a product image search mobile application.



## Image Classification

Build and improve custom image-classification models.



## App Integration

Create and integrate a custom image classifier into a working application.

# The Toolkit Behind the Internship



## Python

Core programming language



## TensorFlow & Keras

Model building & training



## TensorFlow Lite

On-device deployment



## Model Maker

Custom model training



## CNNs

Image recognition



## Object Detection

Pretrained & custom detectors



## Product Image Search

Visual search integration



## Image Classification

Custom classifiers

# From Trained Model to Mobile Application



## Model Layer

CNN built & trained in TensorFlow



## Conversion Layer

Converted via TensorFlow Lite & Model Maker



## Detection Layer

Pretrained & custom object detectors



## Search Layer

Product image search, backend integrated



## Classification Layer

Custom image classifier refines results



## Application Layer

Integrated into a mobile application

# The 8-Week Learning Journey

**1** WEEK

## Neural Networks with TensorFlow

CNNs for image recognition & classification

**2** WEEK

## Get Started: Object Detection

Integrate pretrained detectors into mobile apps

**3** WEEK

## Go Further: Object Detection

Custom models with TFLite & Model Maker

**4** WEEK

## Get Started: Product Image Search

On-device object detection for product search

**5** WEEK

## Product Image Search (contd.)

Static images & live camera feed detection

**6** WEEK

## Go Further: Product Image Search

Backend integration for mobile search app

**7** WEEK

## Go Further: Image Classification

Custom classification models

**8** WEEK

## Go Further: Image Classification

Integrate custom classifier into an app



# Skills & Capabilities Gained



Ability to explain and apply first-principles ML to build CNNs using TensorFlow



Practical understanding of object detection with pretrained & custom TFLite models



Hands-on exposure to building a product image search feature with backend integration



Capability to build, evaluate and improve custom image-classification models



Skill to integrate trained, converted TensorFlow Lite models into a mobile application



Completed all learning-pathway modules & badges; cleared the Final Credential Validation with an “Outstanding” grade

## CERTIFICATION

# Successfully Completed



Grade

O – Outstanding



Certificate ID

b6bf5373f7f3e2aac4c3



Student ID

STUb5f5e89003b33b610a7922b



Duration

8 Weeks | June – August 2026





# Thank You

Google AI-ML Virtual Internship | AICTE – EduSkills Program

**Param Agarwal**

Roll No. 2410030879 | IILM University, Greater Noida